

Half-lives and the random nature of radioactive decay

Atomic structure
Paper 1 · Combined and Separate
8 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 2 starred (★) questions first.

	QUESTION	ANSWER	✓
1	Define half-life in terms of activity.	The time taken for the activity (or count rate) of a sample to halve.	☐
2 ★	Define half-life in terms of the number of nuclei.	The time taken for the number of nuclei of the isotope in a sample to halve.	☐
3	Describe what happens to the half-life of an isotope as time passes.	Nothing: it is constant.	☐
4	Describe what happens to the activity of a radioactive sample over time.	It decreases.	☐
5	Describe what happens to the number of unstable nuclei in a radioactive sample over time.	It decreases.	☐
6	State the fraction of radioactive nuclei remaining after two half-lives.	One quarter ($\frac{1}{4}$)	☐
7	State the fraction of radioactive nuclei remaining after three half-lives.	One eighth ($\frac{1}{8}$)	☐
8 ★	Describe how to find the half-life from a graph of activity against time.	Find the time taken for the activity to fall to half its starting value.	☐